

Smart Automated Questions, Less Risks: Using LLMs to Foster Ethical and Social Awareness in Data-Related Projects

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The integration of machine learning and large language models (LLMs) in finance has boosted efficiency but also introduced socio-ethical risks, including bias, lack of transparency, and potential harm to individuals and society. While such issues are recognized in the literature, they are often underestimated in practice due to motivational and cognitive blind spots: employees may focus narrowly on business goals or technical performance while overlooking ethical and social dimensions. To address this gap, we propose a dialogue system that embeds a structured taxonomy of 26 ethical and social dimensions, grouped under six core principles—autonomy, justice, non-harm, beneficence, transparency, and explainability. Implemented as a prototypical LLM-based chatbot within a Swiss financial institution, the system generates context-aware reflection questions that surface potential blind spots and documents deliberations for compliance and post-audit review. Our evaluation shows that the ethical and social (ES) chatbot consistently outperforms general-purpose LLMs and static questionnaires in generating relevant, clear, and insightful questions. Beyond compliance support, the system’s educational mode can strengthen data literacy and governance processes, offering a practical path toward more ethically aware AI adoption in organizational settings.

Additional Key Words and Phrases: data ethic, dialogue system, social aspects, LLM, ethical decision making

ACM Reference Format:

Yihan Deng, Eleonora Viganò, and Christian Hauser. 2025. Smart Automated Questions, Less Risks: Using LLMs to Foster Ethical and Social Awareness in Data-Related Projects. 1, 1 (February 2025), 25 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

The principal risks linked to the application of machine learning and large language models (LLMs) stem from data biases and intrinsic model limitations. These technical risks can often be mitigated through safeguards such as careful data curation, validation, and monitoring. Legal challenges, by contrast, can be systematically addressed with the support of legal professionals and established risk control frameworks. In financial institutions, however, the *ethical and social (ES) dimensions* of data-driven applications remain markedly underexplored and underdocumented. Their complexity, coupled with a lack of domain-specific expertise, leads to underestimated and/or overlooked socio-ethical risks. As a

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consequence, employees often focus on business goals or technical performance while neglecting the broader impacts on autonomy, fairness, transparency, or accountability, thus creating blind spots in organizational decision-making. To address this gap, we propose a conversational dialogue system tailored for the banking sector that:

- (1) Integrates a structured taxonomy of 26 ES dimensions, derived from established ethical frameworks;
- (2) Implements an LLM-based chatbot that generates context-aware reflection questions to surface ethical blind spots during project evaluation; and
- (3) Provides systematic evaluation, combining system-level metrics, multi-agent simulations, and a user study comparing the chatbot with general-purpose LLMs and static questionnaires.

Together, these contributions demonstrate a practical and scalable approach to embedding ethical and social reflection into financial institutions' data-driven projects.

2 Background

As part of its commitment to fulfilling its social responsibility within the framework of basic public service solutions, our institution subjected data-related financial projects to comprehensive regulatory audits throughout their lifecycle, including the initiation, review, implementation, and operational phases. During the project's initiation phase, multiple business units, along with the legal and IT departments, were actively involved in the review and approval workflows to ensure compliance and alignment with regulatory and organizational standards. Each stakeholder contributed their perspectives on the proposed data-related projects and provided specific comments. Although legal aspects in terms of GDPR have been addressed thoroughly and systematically with GRC-Governance systems, we noted that a comprehensive evaluation addressing ethical and social considerations of the data-related financial projects was notably absent.

Meanwhile, as the emerging field of data and digital ethics continues to evolve, identifying specific domain experts to conduct ethical and social evaluations for all data-related projects has become increasingly challenging, while stakeholders consistently require guidance and insights on ethical and social issues during the project initialization phase. Simultaneously, the most widespread decision-aid tools used by companies for helping employees identify the ES issues of projects (e.g., checklists and guiding questions) proved to be ineffective ([2]). To overcome these difficulties, we developed a dialog system that incorporates knowledge of data ethics and social aspects and employs a mechanism prompting moral reasoning that proved to be effective ([2]). The system supports and streamlines business processes effectively by assisting employees in the socio-ethical assessment of projects. The knowledge of data ethics and social aspects incorporated in the system stems from a framework of data-related ethical and social issues - the HIEDE Framework - that was initialized and developed in the context of internal project with research partners. In this work, we present the architecture design and prototypical implementation of this framework within a financial institution and the experiments proving the efficacy of the dialog system using this framework.

3 Related Work

The ethical and social dimensions of data-related projects have garnered significant scholarly attention, reflecting the complexities and challenges inherent in the data-driven landscape ([9, 23]). A comprehensive understanding of these aspects is crucial for the responsible development and implementation of data-related initiatives.

3.1 Ethical Considerations

Ethical concerns in data projects often revolve around issues of privacy, consent, and the potential for discriminatory biases. A systematic literature review by Saltz et al. (2019) [18] identifies major ethical themes in data science, emphasizing the need for a foundational body of knowledge to navigate ethical dilemmas. Similarly, Olteanu et al. (2019) [16] discuss the ethical boundaries in social data usage, highlighting the risks of biases and methodological pitfalls.

3.2 Social Implications

Social considerations involve the impact of data projects on societal structures, including issues of discrimination, equity, and public trust. The Data Ethics Decision Aid (DEDA), as discussed by Franzke et al. [10], provides a framework for reviewing government data projects, considering their social impact and the embedded values in data-driven public management. To address these multifaceted challenges, integrated frameworks have been developed. The DEDA is one of such frameworks that facilitates ethical inquiry into AI and data projects, promoting a conversational approach to assess social impacts and responsibilities. In conclusion, the literature provided frameworks for assessing the ethical and social aspects of data-related projects, but no evidence of the efficacy of decisions-aids using them.

3.3 Blind Spots in Socio-Ethical Assessment and Current Decision-Aids

Research demonstrates that ES issues and consequent unethical behavior often result from actions that individuals do not realize are morally wrong [4, 19]. In other words, individuals commit unintended unethical actions due to ethical blind spots. The latter are due to implicit cognitive and motivational biases that prevent individuals from acknowledging the ethical dimensions of a given situation or decision. Therefore, in business, failing to address ethical blind spots in a project means risking making unintentional unethical decisions, thus failing to address the ES issues of the project. ([2]).

3.4 Selection of large language model for socio-ethical assessment of projects

Dillon et al. [6] conducted a Moral Turing Test comparing GPT-4o and a renowned ethics columnist (“The Ethicist”) in various ethical alignment tasks—where ethical alignment refers to the extent to which an LLM’s moral judgments and justifications align with human moral intuitions. Their study finds that GPT-4o’s advice was perceived as more moral, trustworthy, thoughtful, and accurate than that of a well-known human ethicist, with participants valuing not only the moral judgments but also the justifications provided. These results provide compelling evidence for the potential of LLMs in tasks involving moral reasoning and ethical alignment. However, further evaluation—especially of open-source LLMs—is necessary to ensure comparable performance without dependence on proprietary, cloud-based systems. This is particularly critical in banking, where internally deployed, auditable systems may be required for moral case assessments. Building on recent philosophical critiques that question the systematicity of truth in normative domains such as ethics and morality [17], we acknowledge the challenges in relying on LLMs for autonomous moral question reasoning and generation. Our chatbot, therefore, emphasizes human-in-the-loop design and context-aware deliberation to mitigate the risks of asystematic ethical outputs. In particular, the ethical conversational interface supports interactive dialogue, enabling users to explore ethical trade-offs through back-and-forth questioning. It also incorporates the loop of justification prompts and human override mechanisms that ensure users can review, challenge, or revise the model’s ethical question generation.

4 Harmonization of Socio-Ethical Frameworks

4.1 Understanding and Considering ES issues in Decision-Making

In business contexts, ethical and social (ES) issues refer to the broader impacts of organizational activities that extend beyond profit maximization. When considered alongside legal concerns, they form the framework of ELSI (Ethical, Legal, and Social Implications), a concept originally developed in relation to large-scale scientific endeavors such as the Human Genome Project ([12]; [15]; [7]). When examining ethical issues, businesses must address questions of right and wrong that go beyond legal requirements. These questions are founded on the moral principles and values of individuals, organizations, communities, etc. Moral principles are norms that guide one's action, such as the principle of beneficence, which requires people to do good to others. Moral values are what individuals or institutions consider a goal worthy of achievement; the value of honesty, for instance, is the commitment to be truthful in words and actions. Social issues are also founded on moral principles and values, but focus on the broader impacts on communities, society, and social structures. Social issues in the business domain regard the company's relationship with broader society including, for instance, environmental responsibility and economic inequality.

4.2 Complexity of ES Management and the support of decision-aids

The answers to questions of right and wrong cannot be derived by applying an algorithm, nor from codified legal requirements providing clear, mandatory boundaries for business conduct. They rather require moral reasoning - the reflection on how one ought to act based on impartial, logically grounded arguments [8]. Moral reasoning in managing ES issues is required mainly for three reasons. First, equally valid moral values and principles are many and may conflict with each other, creating dilemmas. Second, cultural differences also bring about differences in the importance given to some principles over other ones, with the consequence that in multicultural contexts, an action assessed as morally required in one culture may be viewed as inappropriate in another. Third, the answer to what is right to do may vary greatly on the basis of the specific situation and interpersonal dynamics of the individuals involved. Thus, managing ES issues often involves difficult trade-offs and requires careful consideration of multiple perspectives. Moreover, human cognitive and motivational biases add a further level of complexity. As seen, they create blind spots that prevent individuals from recognizing the ES dimensions of a project. As they mostly operate outside conscious awareness, they make people think that they are objective in their assessments. A decision-aid tool that supports employees in the socio-ethical assessment of their projects must therefore aim to reduce users' blind spots. Baader et al. [2] found that reflection questions prompting users to explicitly consider ES risk are effective in reducing their ethical blind spots. To make our decision-aid tool effective, we thus employed Baader et al.'s reflection questions to help employees evaluate the full scope of their operational impact on stakeholders and society.

4.3 Application of the Socio-Ethical Framework

According to current research, the best way to address the ES topics in data-based financial projects is firstly the establishment of institutional ES framework. We have therefore adopted the HIEDE Framework. The latter provides a structured way to categorize and assess relevant types of ethical and social issues that can arise in data-related projects by arranging them in six high-level categories and 20 subcategories. The six categories consist of four ethical principles (autonomy, justice, non-maleficence (prevention of harm) and beneficence) and two ethical *and* epistemic principles specific to projects with data and AI systems (transparency and explainability). As each four ethical principles has internal distinctions that are relevant for a socio-ethical assessment of projects, the taxonomy comprises a total of 20

subcategories that articulate and specify the four ethical principles. Therefore, the proposed taxonomy comprises a total of 26 ethical dimensions (see figure 1). The HIEDE Taxonomy is the synthesis of four established ethical frameworks: three elaborated by academic scholars - ATE (Anticipatory Technology Ethics) [5], EIA (Ethical Impact Assessment of information technology) [24], ETICA (Ethical Issues of emerging ICT Applications) [21, 22] and one elaborated by the High-Level Expert Group on AI appointed by the European Commission - ALTAI from the Ethics Guidelines for Trustworthy Artificial Intelligence [11]. To extend the internal legal template in risk governance software, the HIEDE Framework has been harmonized with the Swiss Post digital ethics scala [1].

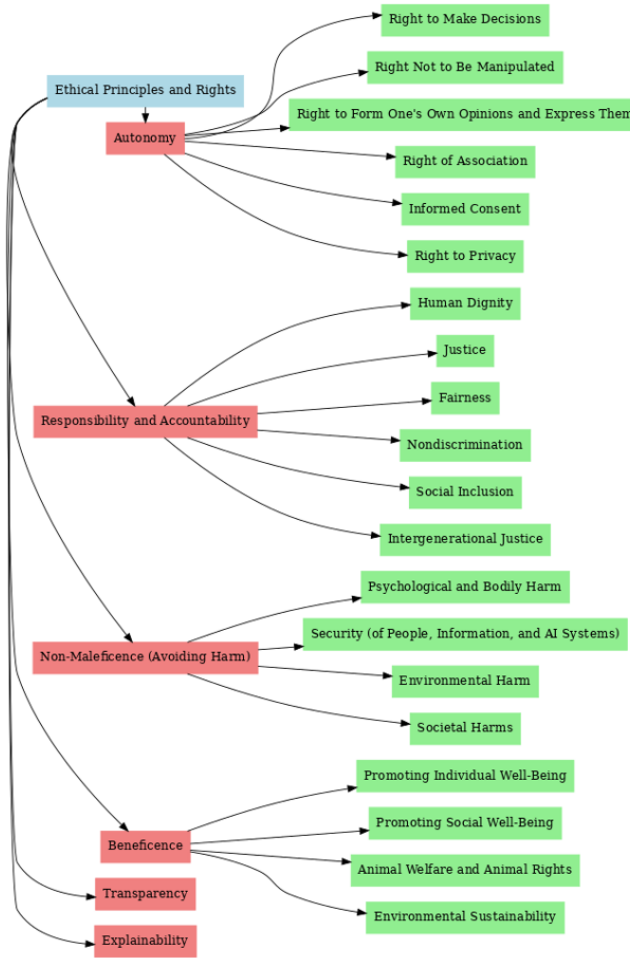


Fig. 1. Proposed Digital ethic-social frameworks

5 Mathematical Definition of ethical based reflective question generation system

Reflective Question Generation for Data-Driven Project Evaluation

In order to rigorously describe the process of generating context-aware reflection questions, we introduce a formal mathematical framework. This formalization serves two purposes:

- (1) It specifies the logical structure of our reflective question generation (RQG) pipeline, ensuring reproducibility and allowing for systematic evaluation.
- (2) It establishes clear notations for the components involved, including ethical aspects, dialogue states, and agent actions.

Let:

- $d \in \mathcal{D}$ be the description of a data-driven project, including goals, methods, data sources, and expected outcomes.
- $\mathcal{R} = \{r_1, r_2, \dots, r_6\}$ be the set of six ES reflection dimensions:
 - r_1 : Autonomy
 - r_2 : Responsibility and Accountability
 - r_3 : Non Maleficence
 - r_4 : Beneficence
 - r_5 : Transparency
 - r_6 : Explainability

Embedding and Representation

Let $e_d = \phi(d)$ be the semantic embedding of the project description using encoder ϕ , capturing the ethical context and technical details.

Dimension-Specific Prompting

For each ethical dimension $r_i \in \mathcal{R}$, define a prompt function p_i that generates a context-aware ES reflection prompt:

$$p_i(e_d) = \text{Prompt for ethical dimension } r_i \text{ based on } e_d$$

Example prompts:

- $p_1(e_d)$: "How does the project ensure user autonomy in decision-making and data usage?"
- $p_2(e_d)$: "Who is accountable for the outcomes of the model, and how is responsibility distributed?"
- $p_3(e_d)$: "What measures are in place to prevent harm to individuals or communities?"
- $p_4(e_d)$: "In what ways does the project aim to benefit stakeholders or society at large?"
- $p_5(e_d)$: "How transparent are the data sources, model decisions, and system limitations?"
- $p_6(e_d)$: "Can the system's decisions be explained in a way that is understandable to non-experts?"

Question Generation

Let f_{RQG} be a reflective question generator model. Then the set of generated ES reflection questions is:

$$\mathcal{Q}_{\text{reflect}} = \bigcup_{i=1}^6 f_{\text{RQG}}(p_i(e_d))$$

Evaluation and Filtering

Let $s(q, a_i)$ be a scoring function that evaluates each generated question q for aspect a_i along six dimensions used in our human studies:

- **Relevance** $R(q, a_i)$: Appropriateness of q with respect to ethical aspect a_i (1–5 Likert).
- **Clarity** $C(q)$: Degree to which q is understandable and clearly phrased (1–5 Likert).
- **Insightfulness** $I(q, a_i)$: Capacity of q to provoke deep or expert-level reflection (1–5 Likert).
- **Human-likeness** $H(q)$: Similarity of q to what a human expert might ask (1–5 Likert).
- **Attention-grabbing** $A(q)$: Likelihood that q challenges or engages the user (1–5 Likert).
- **Coverage** $\text{Cov}(Q)$: Balance and variety across aspects in a set of questions Q (heuristic score in $[0, 1]$).

Let τ be a threshold score. Then the final set of ethically reflective questions is:

$$Q_{\text{final}} = \{q \in Q_{\text{reflect}} \mid s(q, r_i) \geq \tau \text{ for some } r_i \in \mathcal{R}\}$$

Question Generation Agent (QGA)

Let:

- $x \in \mathcal{X}$ be the input context (e.g., user query, document, or dialogue history),
- $Q_1 = \{q_1, q_2, \dots, q_n\}$ be the set of generated questions.

The QGA maps context to questions:

$$Q_1 = f_{\text{QGA}}(x)$$

where f_{QGA} is a generative model trained to produce relevant and ethically sound questions.

Criticizer and Optimizer Agent (COA)

This agent evaluates and improves the generated questions.

Let:

- $s(q)$ be a scoring function that evaluates question quality based on ethical criteria,
- $Q_2 = \{q'_1, q'_2, \dots, q'_m\}$ be the set of optimized questions.

Then:

$$Q_2 = \{g(q) \mid q \in Q_1, s(q) < \tau\}$$

where:

- $g(q)$ is an optimization function that revises q ,
- τ is a threshold for acceptable ethical quality.

Selector and Voter Agent (SVA)

This agent selects the best questions from both sets Q_1 and Q_2 .

Let:

- $Q = Q_1 \cup Q_2$,
- $v_i(q)$ be the vote or score from agent i for question q ,
- $V(q) = \sum_{i=1}^k v_i(q)$ be the aggregated score from k agents.

Then the final selected set is:

$$Q_{\text{final}} = \{q \in Q \mid V(q) \geq \theta\}$$

where θ is a voting threshold.

System Flow Summary

$$\text{Input} \rightarrow \text{QGA} \rightarrow Q_1$$

$$Q_1 \rightarrow \text{COA} \rightarrow Q_2$$

$$Q_1 \cup Q_2 \rightarrow \text{SVA} \rightarrow Q_{\text{final}}$$

6 System implementation toward participatory ES-Informed Decision-Making

Given the multifaceted complexity of managing ES implications in business, there is a growing need for systems that not only encode ES knowledge but also actively engage users in reflective decision-making. Professionals are increasingly expected to make ethically sound decisions in data-driven projects—often under uncertainty, time pressure, and conflicting interests. However, as seen, empirical research highlights that currently ES risks are frequently underestimated or overlooked due to ethical blind spots.

To address this challenge, we propose the development of an interactive dialogue system that embeds ES principles and supports users through targeted reflection questions. Inspired by self-regulatory approaches, such a system can prompt users to evaluate both their current decision context (the "is" state) and the ethical, legal, or normative standards they aim to uphold (the "ought" state). By surfacing discrepancies between intent and action, and by guiding users through structured moral reflection, the system can help improve ethical awareness and decision quality.

Crucially, this approach moves beyond static checklists and embraces conversational scaffolding to help users explore ethical blind spots and thus gain ethical awareness, clarify competing values, and build moral competence. Through simple but powerful heuristics—structured as yes/no prompts or value-aligned diagnostic questions—the dialogue system serves not only as a decision aid but also as a tool for moral education and value-sensitive sense-making. In doing so, it aligns with recent findings that lightweight, cognitively accessible tools can outperform complex ethical frameworks, especially under uncertainty [2]. This integration of dialogue, reflection, and ethical prompting represents a promising step toward operationalization ES issues in everyday decision contexts.

Based on the proposed HIEDE Taxonomy, we have implemented a prototypical agent-based dialogue system (see figure 2), designed to support internal coordination and ethical review of data-related projects through reflective conversations. The system leverages multiple on-premise LLMs for language understanding and dialogue management, generating reflective prompts aligned with predefined ethical criteria. The underlying ethical knowledge structures can be updated at runtime via an interactive knowledge editing interface. Additionally, the system integrates documentation retrieval and statistical analysis of historical chat logs, enabling report generation and supporting post-audit review of projects. The system is structured with the following components:

6.1 Agent based Conversational Interface

As illustrated in Figure 2, Component 1 (highlighted in red) serves as the central coordinator of the HIEDE ES Bot system. It is implemented following the LangGraph ReAct agent architecture, which handles user queries, interfaces

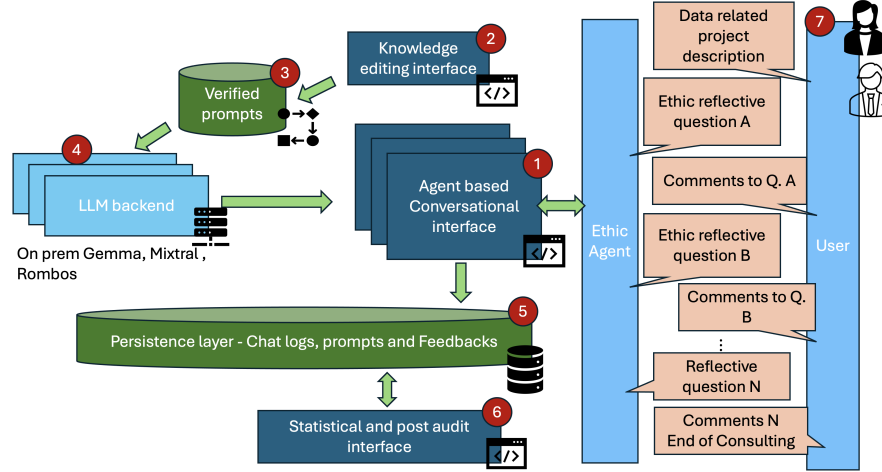


Fig. 2. System Architecture of HIEDE-PF ES Chatbot

with back-end LLMs, and initiates the response generation process based on a predefined ES schema (see figure 6 in Appendices). The agent also incorporates a memory store that maintains the full conversational context in a structured format (figure 3), enabling context-aware interaction and facilitating subsequent retrieval for audit or analysis purposes (see figure 8 in Appendices).

6.2 User-Bot Dialogue Flow

Component 7 in Figure 2 manages the structured consultation flow between the user and the chatbot for ES issue support. The interaction begins with the user submitting a project or data-related description, upon which the chatbot generates targeted, aspect-specific reflection questions grounded in six predefined ES dimensions. For each ES aspect, the user can configure the intensity level of questioning, choosing to receive between one and three reflective prompts. These questions are formulated within the broader context of the project and are designed to elicit feedback from stakeholders such as product owners or customer success managers. Once all six ES aspects have been addressed and commented on, the consultation session is automatically closed. All questions and user responses are logged for transparency, traceability, and post-session reporting.

6.3 LLM backend

At the core of the system, we have deployed a Hugging Face Text Generation Inference (TGI) server within our on-premise GPU cluster, hosting multiple scalable LLMs, including Mixtral 8x7B, Gemma 3 24B, and Rombos 2.4 14B. This setup enables low-latency processing of user prompts and dynamically generates context-aware, ES aligned responses. To ensure consistency in output structure and response quality across different models, we utilize a cross-LLM prompt templating strategy. Each model generates reflective questions based on the structured socio-ethical taxonomy, drawing from the user's project description and session history. As illustrated in Figure ??, prompts are designed to address key ES dimensions, supporting the reduction of potential blind spots through targeted questioning. Figure ?? shows an example of the generated questions in response to a fictitious case description.

6.4 Knowledge Author

Component 2 (Knowledge Author / Aspect Editor) enables runtime adaptation of ES focus areas by allowing users to edit, refine, or extend the predefined ES aspects used by the system. This interactive interface supports domain experts in tailoring the ES taxonomy to evolving project contexts or organizational priorities without redeploying the system (see figure 9 in Appendices). Component 3 (Verified Prompt Module) operates in a feedback-driven evaluation loop, providing benchmark-based assessments of prompt effectiveness across the supported LLMs. It identifies and maintains a set of optimized prompts that maximize response relevance, clarity, and consistency for each model, ensuring the system’s reflective outputs remain both efficient and robust across diverse deployments. The HIEDE Taxonomy serves as the foundational knowledge base, encompassing the ES dimensions presented in Figure 1. This structured knowledge base enables the chat system to generate targeted and domain-specific questions and facilitate the prompt generation for LLM finetuning.

6.5 Context Awareness and Session Control

The dialogue system is equipped with advanced context-awareness through real time reprompting, allowing it to maintain relevant information across multiple exchanges during a session [20]. Session-based context control ensures that the dialogue remains coherent and that subsequent interactions are informed by prior exchanges. Specifically, the user’s responses during the dialogue will be analyzed and dynamically incorporated into the prompt at runtime, enabling the chat to adapt to the ongoing session.

[INST]

Instruction: You are a helpful ethical evaluation assistant named HIEDE-Ethical chatbot.

Language: German

Context: Project description, chat history, and ethical framework

Your task is to generate reflection questions in German based on the provided project description, ethical framework, and previous dialogue history.

DESCRIPTION

Here is the Project description:

{description}

Framework

Here are the predefined ethical aspects from different levels (categories) to guide the questions:

{framework}

CHAT

Here is the chat history to help you grasp the user opinion:

{chat_history}

TASK

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For each ethical aspect listed in the framework, generate exactly ONE reflection question in German that will help the user critically evaluate this aspect of their project.

Along with each question, provide a short explanation (1–2 sentences) why this question is relevant.

RESPONSE FORMAT

- **Aspekt**: [ethical aspect name]

Frage: [reflection question in German]

Erklärung: [short explanation why the question is important]

[/INST]

6.6 Voting of reflection question for better ES judgments through Agent to Agent Chat

The Agent-to-Agent Chat is a multi-agent reflective approach where several LLM agents debate diverse ES perspectives and a final “judge” synthesizes their conclusions. This approach draws on robust research showing its value for nuanced reasoning and socio-ethical deliberation. For instance, adversarial multi-agent evaluation frameworks (e.g., advocates, judge, and jury roles in courtroom-inspired debates) have demonstrated superior performance in error reduction and alignment with human judgments compared to single-model evaluation [3]. In bioethics, ADEPT—a system simulating panels of AI agents representing ethical frameworks such as deontology and consequentialism has been successfully applied to ventilator-allocation dilemmas, revealing how panel composition influences reasoning paths even when final conclusions match [27]. Other work on hallucination-mitigation combines adversarial debate and voting among agents to cross-verify outputs and reduce misinformation [25]. Together, these studies highlight key benefits—enhanced diversity of viewpoints, improved factual grounding through cross-agent critique, error checking, and transparent final synthesis by a judge agent—making structured LLM debate both practical and well-grounded in state-of-the-art AI research. We will evaluate our multi-agent system for generating reflective questions using four complementary metrics: Debate Accuracy and Consensus Alignment, Error and Hallucination Reduction Metrics, MATEval-style Multi-Agent Evaluation, and Agent-Level Contribution and Role Analysis. Debate Accuracy and Consensus Alignment measures how often the final judge agent’s output aligns with human judgments on relevance, reflection, and ethical depth. Error and Hallucination Reduction Metrics tracks the decrease in factual mistakes using adversarial debate and voting mechanisms, quantified via hallucination rates and accuracy improvements. MATEval-style Multi-Agent Evaluation leverages a structured discussion framework that incorporates chain-of-thought and feedback loops [13], evaluated using semantic metrics such as BERTScore [26] and human correlation. Agent-Level Contribution and Role Analysis examine the unique contributions of each agent for novelty and frequency of reflective questions to assess collaborative efficiency. Combining these metrics will yield a robust, multidimensional view of system performance, spanning human alignment, factual reliability, discourse structure, and individual agent effectiveness [13].

7 Adapting Emotional Support Strategies to Enhance Socio-Ethical Reflection

Inspired by the design of emotionally supportive dialogue systems [14], we extend our ES reflection framework by incorporating emotional scaffolding mechanisms originally designed to reduce users’ emotional distress. Below, we describe how the ESC (Emotional Support Conversation) framework and its annotated strategies can improve ES awareness in data-driven contexts.

7.1 Mapping ESC Stages to Socio-Ethical Reflection

The ESC framework defines a three-stage structure: *Exploration*, *Comforting*, and *Action*. We adapt these stages to deepen ES deliberation within our multi-agent system:

- **Exploration:** Uncovers latent ES concerns in a project, such as conflicting values or blind spots.
- **Comforting:** Frames ES tensions empathetically, helping users recognize dilemmas without defensiveness.
- **Action:** Suggests resolution strategies or mitigation directions to support informed ES decisions.

This structure ensures that generated questions move beyond binary moral checks and support a reflective trajectory.

7.2 Incorporating Emotional Support Strategies into Prompting

We incorporate the eight annotated support strategies from ESConv [14] into our prompt design to elicit more relatable and nuanced questions. These strategies include:

- **Questioning** – e.g., “What socio-ethical dilemmas might be overlooked in this project?”
- **Reflection of Feelings** – e.g., “Some stakeholders may feel disempowered—how is that addressed?”
- **Self-disclosure** – e.g., “As an auditor, I might also worry about fairness here.”
- **Affirmation and Reassurance** – e.g., “It’s good that fairness is being considered—how can it be strengthened?”
- **Suggestion and Information** – e.g., “Could transparency guidelines help resolve this tension?”

These strategies enrich the tone and depth of questions, fostering more human-aligned interactions.

7.3 Persistence Layer for chatlogs, prompts and feedbacks

The persistence layer (component 5) is implemented using an OpenSearch vector database, which stores all chatbot-generated questions, user responses, and feedback throughout the consultation sessions. This vectorized storage enables semantic retrieval, allowing users to query historical interactions based on content similarity rather than keyword matching. By maintaining a structured and searchable record of ethical dialogue, the system supports longitudinal tracking of ethical considerations and enables automated generation of structured reports at various stages of the project life cycle, including planning, mid-term review, and post-audit evaluation.

7.4 Documentation and Evaluation

Component 6 is dedicated to structured reporting and post-audit evaluation. It systematically documents the full dialogue history—including all reflection questions and user responses—for use in project auditing, compliance tracking, and retrospective analysis. This component also supports performance evaluation by assessing whether the system’s outputs align with predefined ethical and social standards, providing actionable insights for iterative improvement. Through the conversational user interface, users receive tailored ES guidance for their data-related projects, while all stakeholders can access a comprehensive summary and chat protocol, offering an integrated view of the addressed ES aspects to support transparent, data-driven decision-making across the project life cycle.

8 Evaluation

To comprehensively assess the capabilities and robustness of the proposed ES chatbot framework, we adopt a multi-pronged evaluation strategy encompassing both **system-oriented** and **user-oriented** perspectives, complemented by targeted **multi-agent simulation** experiments.

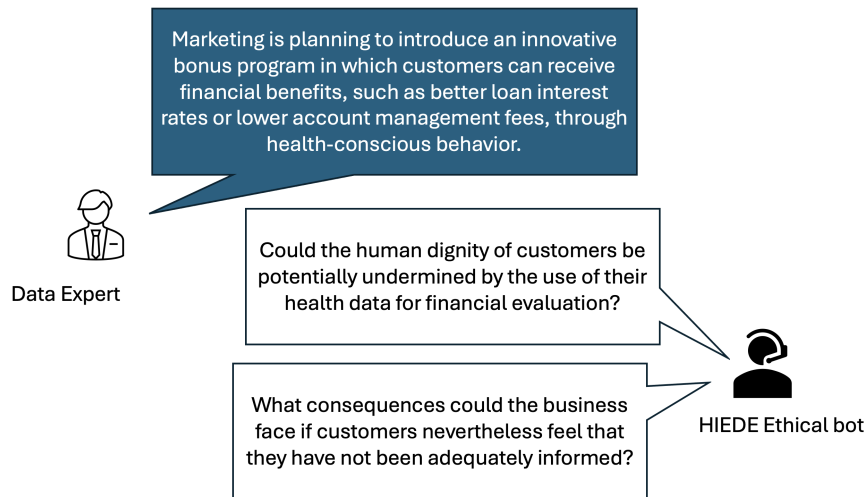


Fig. 3. Reflection Questions (hints) generation through Ethic Bot with input of fictitious unethical case

From a *system-oriented* perspective, we evaluate the chatbot's performance in controlled, reproducible settings using structured ground truth (standard response, chat paths) judged by human experts, that test socio-ethical question generation across single-aspect, multi-aspect, and adversarial user scenarios. Metrics such as category relevance, clarity, insightfulness, human likeness, ES consistency, critical error rate, responsiveness, and aspect coverage as well as attention grabbing provide quantitative insight into the model's ability to uphold ethical principles under varied conversational conditions.

The *user-oriented* evaluation employs a within-subjects user study involving 20 participants from diverse professional backgrounds. Four AI systems—ChatGPT-4o, ChatGPT-5, ES Chatbot, and Questionnaire are compared in their ability to generate reflective questions and support ES analysis of realistic case studies. Participants rate each system across multiple quality dimensions (relevance&depth, clarity&Coherence, usefulness, and user experience 2), enabling direct comparison of general-purpose versus domain-specific AI solutions.

In parallel, a *multi-agent reflection simulation* assesses whether iterative collaboration between specialized agents (question generation, critique/improvement, and selection) can enhance the quality of reflective questions. This setup allows controlled measurement of quality convergence, trade-offs between agent configurations, and the point of diminishing returns for iterative refinement.

Finally, integration considerations and future enhancements—such as expanding the knowledge base, aligning with legal frameworks (e.g., GDPR), and enabling educational modes—are planned to ensure that the validated system can be effectively deployed in real-world data governance and ethics workflows.

8.1 System-oriented Evaluation of question coverage and completeness of ES chatbot

This section reports on a set of system-oriented evaluations that examine the ability of large language models (LLMs) to generate reflective questions aligned with predefined socio-ethical (ES) aspects. The analyses proceed in four stages. First, a comparative study assesses ES awareness and question quality across different LLMs. Second, a multi-agent

Criterion	Description	Rating Scale
Relevance	Appropriateness of the question with respect to the given ethical aspect	1–5 Likert
Clarity	Degree to which the question is understandable and clearly phrased	1–5 Likert
Insightfulness	Capacity of the question to provoke deep or expert-level reflection	1–5 Likert
Human-likeness	Similarity of the question to what a human expert might ask	1–5 Likert
Attention-grabbing	Likelihood that the question challenges or engages the user	1–5 Likert
Coverage	Balance and variety of questions across all ethical aspects	Heuristic score

Table 1. Evaluation Criteria for Reflective Question Quality

simulation investigates how iterative interactions between generator, critic, and selector agents can enhance the clarity, depth, and relevance of generated questions. Third, the study explores optimal strategies for agent configurations and refinement rounds to balance performance with computational efficiency. Finally, an evaluation of emotional feedback mechanisms is conducted to determine their impact on user engagement and ethical clarity. Collectively, these experiments provide a comprehensive assessment of system performance and design considerations for deploying LLMs in ethically guided user interaction contexts.

8.1.1 Comparison of ES awareness between different LLMs. This experiment aims to evaluate and compare the ES awareness and reflective question quality of two LLMs based on their responses to data-driven scenarios. The primary objective is to determine which model produces more suitable, expert-level, and human-like questions that effectively capture user attention while aligning with predefined ES aspects. **Dataset Preparation.** A fixed set of 25 scenarios is curated. Each scenario describes a real-world or synthetic case involving an AI system or decision-making process with potential socio-ethical implications. Scenarios are annotated with one or more relevant ES aspects to guide question generation. **Socio-Ethical Aspects:** The evaluation uses the list of six ES aspects of the socio-ethical taxonomy: (1) Autonomy, (2) Responsibility and Accountability, (3) Non Maleficence, (4) Beneficence, (5) Transparency, and (6) Explainability. **Evaluation Criteria:** The generated questions are evaluated by 5 human raters and 1 prompted LLM (GPT-4o) rater according to several dimensions (Table 1). These include relevance, clarity, insightfulness, human-likeness, attention-grabbing quality, and coverage. Table 2 shows aggregated results with 95% confidence intervals, overall, Gemma 3 24B achieved the highest scores in all dimensions, particularly in clarity (4.46 ± 0.22) and insightfulness (4.52 ± 0.20), indicating strong performance in generating relevant and human-like questions. In contrast, Qwen 2.5 32B showed the lowest ratings across categories, while Mixtral 8x7B consistently performed in between the two models. Statistical significance was assessed using one-way ANOVA for each criterion (Table 3), revealing significant differences across the evaluated models (Mixtral 8x7B, Gemma 3, Qwen 2.5 32B). To evaluate whether the differences in model performance were statistically significant, a one-way ANOVA was conducted for each of the five evaluation metrics. The results in table 3 revealed statistically significant differences across the three LLMs (Mixtral 8x7B, Gemma 3, and Rombos 2.6 24B) for all metrics evaluated.

These results in table 3 confirm that the observed differences in rating distributions among the LLMs are highly statistically significant across all evaluated dimensions, supporting the conclusion that model architecture and design meaningfully affect the ES-related quality of generated content. The evaluation of model performance combines quantitative and qualitative perspectives. Mean scores are computed per criterion and statistical tests (e.g., ANOVA) are

Table 2. Descriptive Statistics of Ethical Question Quality (Mean $\pm$ Std)

Model	Relevance	Clarity	Insightfulness	Human-likeness	Attention-grabbing	Coverage
Gemma 3 24B	4.40 $\pm$ 0.13	4.46 $\pm$ 0.22	4.52 $\pm$ 0.20	4.28 $\pm$ 0.17	4.39 $\pm$ 0.18	1.06 $\pm$ 0.11
Mixtral 8x7B	4.08 $\pm$ 0.18	4.03 $\pm$ 0.17	4.00 $\pm$ 0.22	4.21 $\pm$ 0.16	4.06 $\pm$ 0.21	1.04 $\pm$ 0.13
Qwen 2.5 32B	3.49 $\pm$ 0.21	3.62 $\pm$ 0.16	3.47 $\pm$ 0.16	3.73 $\pm$ 0.19	3.48 $\pm$ 0.22	1.00 $\pm$ 0.02

Table 3. One-way ANOVA Results for Model Comparison with Effect Sizes (η^2)

Criterion	F-statistic	p-value	η^2
Relevance	168.52	6.92×10^{-28}	0.55
Clarity	126.29	2.86×10^{-24}	0.52
Insightfulness	179.49	1.06×10^{-28}	0.58
Human-likeness	76.16	1.71×10^{-18}	0.42
Attention-grabbing	125.45	3.44×10^{-24}	0.55
Coverage	1.76	0.179	0.05

applied. In addition, qualitative feedback from human annotators provides insight into stylistic and thematic differences across models. To complement the ANOVA results, we also report effect sizes to illustrate the practical magnitude of observed differences. For the system-level evaluation, one-way ANOVA effect sizes (η^2) were computed and yielded large values across all criteria (Relevance: $\eta^2 = 0.55$, Clarity: $\eta^2 = 0.52$, Insightfulness: $\eta^2 = 0.58$, Human-likeness: $\eta^2 = 0.42$, Attention-grabbing: $\eta^2 = 0.55$), confirming that the model factor accounts for a substantial proportion of the variance in ratings. ($d = 0.4$ – 0.6 across criteria). These effect sizes support the reliability of the findings by demonstrating that the improvements are not only statistically significant, but also practically meaningful.

8.1.2 Multi-agent simulation for reflection question generation. We simulate a multi-agent reflection system using ReAct agents with the goal of iteratively improving the quality of reflective questions across multiple ES dimensions (see Figure 4). The simulation evaluates **prompt effectiveness**, the **number of required turns**, and **mechanisms for quality assurance and convergence**. The system includes three core agent types: **Agent 1 (Generator)** generates one reflection question per ES aspect (e.g., autonomy, beneficence); **Agent 2 (Critic and Improver)** evaluates each question using predefined criteria such as clarity, relevance, and depth, and provides improved versions; **Agent 3 (Selector)** compares the original and revised questions, selects the better one, and optionally justifies the decision for traceability.

The simulation follows a 5-step protocol: (1) define six socio-ethical aspects: Autonomy, Responsibility & Accountability, Non Maleficence, Beneficence, Transparency and Explainability; (2) select a data-driven case such as a predictive personal pension model trained on 10 years of transaction data; (3) apply structured prompts to each agent (e.g., Agent 1: “Generate one reflective question per aspect.” Agent 2: “Rate each question on clarity, relevance, and depth (1–5) and provide an improved version.” Agent 3: “Select the question that better supports ES reflection and justify your choice.”); (4) implement iterative rounds where Agent 1 refines questions based on Agent 3’s selections, looping back through Agents 2 and 3; (5) measure improvement using a scoring rubric, embedding similarity, and optionally a meta-agent judge (see figure 7 in Appendices).

The system’s turn-based logic is described in the following pseudocode:

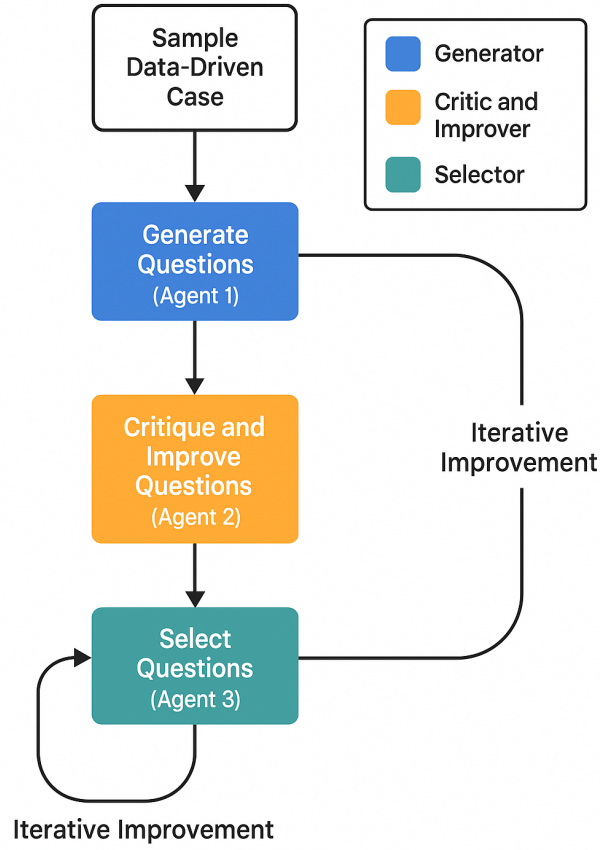


Fig. 4. Multi-agent flowchart for 3 agents simulation

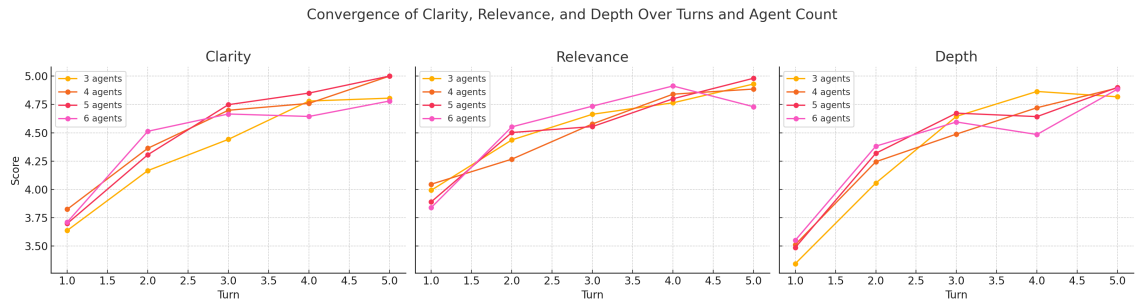


Fig. 5. Convergence of clarity, Relevance and Depth Over turns and Agent count

```

833 for turn in range(max_turns):
834     agent1_questions = agent1_generate(aspects, case)
835     agent2_review = agent2_critique(agent1_questions)
836     agent3_selection = agent3_choose(agent1_questions, agent2_review)
837     log_history(turn, agent1_questions, agent2_review, agent3_selection)
838     if convergence(agent3_selection, history):
839         break
840
841

```

8.1.3 *Insights and Optimal Strategy of multi-agent question generation.* As is shown in figure 5, increasing the number of agents generally leads to higher initial scores, with the most notable improvement observed for **Depth**, which benefits from more specialized feedback. However, the gains begin to plateau by **Turn 3–4**, particularly for **Clarity** and **Relevance**, indicating diminishing returns beyond this point. Configurations involving six agents tend to outperform others in terms of Depth, underscoring the value of aspect-specific expertise. In contrast, a configuration with three agents is typically sufficient for achieving strong performance in Clarity and Relevance, although it may be less effective in producing deep ES insights.

Based on these observations, the optimal strategy involves conducting approximately three iterative rounds, which is generally sufficient to achieve substantial quality improvements without incurring unnecessary computational cost. In practice, this suggests that organizational deployments should limit iterative refinement to three rounds for efficiency without sacrificing quality. For lightweight setups or initial brainstorming phases, employing three agents is recommended. When the objective is to generate in-depth, high-quality ES reflection across multiple dimensions, a configuration of five to six agents provides the best balance between performance and resource requirements.

8.1.4 *Evaluation of emotional feedbacks.* The emotional support system in [14] evaluates success via emotion intensity reduction. We adopt a parallel structure by introducing:

- **Reflection depth metrics** – Tracking how effectively a question prompts user reasoning.
- **Insight progression feedback** – Letting users rate the usefulness of reflective questions.
- **Session-phase analysis** – Segmenting conversations to track evolution of ethical clarity.

These mechanisms ensure that socio-ethical questioning is not only relevant but cognitively constructive. Table 4

Table 4. Impact of Emotional Support System on Evaluation Feedbacks

Metric	With Emotional Support System	Without Emotional Support System
Reflection Depth (avg. score)	4.5	3.8
Insight Progression (user-rated usefulness)	4.4	3.5
Ethical Clarity Evolution (session coherence)	4.6	3.9
User Engagement (turns per session)	12	8
User Satisfaction (Likert 1–5)	4.7	3.9

illustrates the advantages of integrating an emotional support system into the ES Chatbot. Across all key dimensions of user interaction and ES reflection, the emotionally aware version outperforms the baseline system. Specifically, it achieves higher scores in reflection depth (4.5 vs. 3.8) and insight progression (4.4 vs. 3.5), indicating that users found the reflective questions more meaningful and thought-provoking. ES clarity also improved (4.6 vs. 3.9), suggesting that

conversations became more coherent and ethically grounded over time. Furthermore, the system fostered greater user engagement, with an average of 12 turns per session compared to 8 in the control setup. Overall satisfaction was notably higher (4.7 vs. 3.9), confirming that users perceived the emotionally responsive chatbot as more supportive and valuable in facilitating ES reasoning. The results of the experiment confirm that the expected outcomes have been effectively fulfilled. The comparative analysis of the three LLMs—Gemma 3 24B, Mixtral 8x7B, and Qwen 2.5 32B—revealed distinct strengths and weaknesses across the six predefined ethical aspects. Aggregate scores across multiple quality dimensions, including clarity, relevance, and insightfulness, provided robust rankings and highlighted Gemma 3 24B as the most consistent performer. Representative examples from the generated questions further illustrated how each model varies in ethical framing and depth of reflection. Drawing on both system-level performance metrics and user evaluation results, the findings provide concrete guidance on how each model can be deployed in ethically guided user-interaction contexts, thereby confirming the validity of the experimental design and demonstrating that the original objectives have been achieved. The combination of system-oriented metrics and user evaluations provides clear guidance on how each model should be deployed in ethically guided user interactions. These findings confirm the validity of the experimental design and demonstrate that the original objectives have been achieved.

8.2 User study for ES Chatbot vs ChatGPT-4o vs ChatGPT-5 vs Questionnaire

The primary objective of this study is to evaluate and compare the capabilities of three AI systems—ChatGPT-4o, ChatGPT-5, ES Chatbot, and Questionnaire—in generating ES reflective questions and supporting ES evaluations of data-driven projects. These systems were chosen for their distinct design philosophies: ChatGPT-4o/5 as a state-of-the-art general-purpose LLM, ES Chatbot as a specialized ES-focused conversational agent, and Questionnaire as a structured questionnaire-based tool. By comparing them, we aim to determine how general-purpose and domain-specific AI differ in fostering ES awareness and reflection.

A total of 20 participants (11 female, 9 male) were recruited, aged 24–58 years (mean: 36.7, SD: 8.4), representing diverse professional backgrounds including compliance (6), analytics (9), and marketing (5). At least 70% had prior experience in data science, AI ethics, or digital technology policy, ensuring informed evaluation while maintaining heterogeneity of perspectives.

Participants were presented with three real-world-inspired case studies: (1) AI in Hiring, focusing on algorithmic bias and transparency; (2) Predictive Policing, addressing fairness, privacy, and accountability; and (3) Health Data Analytics in financial domain, covering informed consent and beneficence. All participants received the same scenarios, which were presented with standardized background descriptions and no system-specific cues.

A within-subjects design was employed so that each participant interacted with all four systems. The order of system presentation was randomized for each participant to mitigate order effects. Each participant engaged with the same case studies across all systems.

For each case study and system:

- (1) Participants read the case description.
- (2) They interacted with the assigned system to generate ES reflective questions.
- (3) They reviewed the system’s output and used it to formulate an ES evaluation of the project.
- (4) Interaction time per system was capped at 10 minutes per case.
- (5) Clarification questions to the experimenter were permitted only for procedural issues, not content guidance.

After interacting with the systems, participants rated them based on several evaluation criteria:

- **Relevance and Depth:** Extent to which questions addressed key ES issues and encouraged deep reflection.
- **Clarity and Coherence:** Ease of understanding and logical flow of the questions.
- **Usefulness for Ethical Analysis:** Perceived value in supporting ES decision-making.
- **Overall User Experience:** Satisfaction with interacting with the system.

Likert scales (1 = strongly disagree, 5 = strongly agree) were used for quantitative ratings, with definitions of each criterion provided in the participant guide. Quantitative data were collected using an online survey platform, which also embedded the open-ended qualitative feedback form. Participants provided a final comparative ranking of the three systems.

Open-ended feedback was analyzed using thematic coding. Two independent coders identified recurring themes related to strengths, weaknesses, and user expectations. Disagreements were resolved through discussion, and Cohen's kappa was computed to measure inter-rater reliability ($\kappa = 0.84$). Repeated-measures ANOVA was used to test for differences in quantitative ratings between systems, followed by Bonferroni-corrected pairwise comparisons. Qualitative themes were summarized to contextualize quantitative results.

- Statistical analysis was conducted on the quantitative ratings to identify significant differences.
- Thematic analysis was applied to qualitative feedback to extract insights into user perceptions and system performance.

The two sets of evaluation criteria serve complementary purposes. The system-oriented metrics focus on robustness and reliability of the chatbot itself, measuring whether it consistently upholds socio-ethical standards across varied conversational conditions. In contrast, the user study criteria (clarity, relevance, usefulness, and overall user experience) capture a human-centered perspective, emphasizing how participants perceive the quality and practical value of the system's reflective questions. By combining these viewpoints, the evaluation balances technical consistency with subjective usability, ensuring that the chatbot is not only ethically reliable in principle but also meaningful and engaging for end users in practice.

Table 5. User Study Ratings Summary (Mean $\pm$ Std across 20 participants and 3 cases)

Model	Relevance & Depth	Clarity & Coherence	Usefulness	User Experience
ES Chatbot	4.43 $\pm$ 0.29	4.46 $\pm$ 0.31	4.31 $\pm$ 0.27	4.43 $\pm$ 0.31
ChatGPT-4o	4.31 $\pm$ 0.29	4.41 $\pm$ 0.30	4.11 $\pm$ 0.27	4.19 $\pm$ 0.29
ChatGPT-5	4.35 $\pm$ 0.25	4.01 $\pm$ 0.30	4.06 $\pm$ 0.28	4.01 $\pm$ 0.23
Questionnaire	3.48 $\pm$ 0.32	3.60 $\pm$ 0.28	3.29 $\pm$ 0.30	3.68 $\pm$ 0.29

Table 6. Pairwise Comparisons of Systems (Cohen's d Effect Sizes). Strong effects > 0.8 , moderate ≈ 0.5 , weak ≈ 0.2 .

Comparison	Relevance & Depth	Clarity & Coherence	Usefulness	User Experience
ES Chatbot vs. Questionnaire	3.11 (strong)	2.96 (strong)	3.46 (strong)	2.50 (strong)
ES Chatbot vs. ChatGPT-4o	0.41 (weak)	0.55 (moderate)	0.71 (moderate)	0.82 (moderate)
ChatGPT-4o vs. Questionnaire	2.72 (strong)	2.79 (strong)	2.87 (strong)	1.76 (strong)
ChatGPT-5 vs. Questionnaire	3.03 (strong)	1.41 (strong)	2.65 (strong)	1.26 (strong)
ChatGPT-5 vs. ES Chatbot	0.30 (weak)	0.65 (moderate)	0.88 (strong)	0.95 (strong)
ChatGPT-5 vs. ChatGPT-4o	0.15 (weak)	1.33 (strong)	0.18 (weak)	0.69 (moderate)

As shown in Table 5 and Table 6, the effect sizes confirm that the observed differences are not only statistically significant but also practically meaningful. The ES Chatbot demonstrates consistently large improvements over the Questionnaire across all criteria (Cohen’s $d > 2.5$), underscoring its substantial advantage in stimulating relevant and useful reflections. Compared to ChatGPT-4o, the ES Chatbot exhibits moderate gains, particularly in Clarity and User Experience ($d \approx 0.5$ – 0.8), highlighting its strength in producing coherent, user-aligned dialogue. ChatGPT-4o itself also shows strong effects over the Questionnaire ($d \approx 1.7$ – 2.9), confirming that conversational systems in general markedly outperform static questionnaires. Against ChatGPT-5, the ES Chatbot retains an advantage in Clarity and User Experience (moderate to strong effects), while ChatGPT-5 achieves higher scores in Relevance and Depth. These findings highlight that the ES Chatbot’s benefits are both robust and targeted: it not only secures broad improvements over static tools but also delivers superior coherence and user-centered interaction compared to more general-purpose LLMs.

Table 7. Final System Preferences from 20 Participants

Model	Participants Preferring
ES Chatbot	11
ChatGPT-4o	5
ChatGPT-5	3
Questionnaire	1

Summary of User-Oriented Evaluation: Across 20 participants and 3 data-driven cases, the ES Chatbot consistently received the highest ratings in three of the four evaluation dimensions: **Clarity & Coherence**, **Usefulness**, and **User Experience**. It achieved an average rating above 4.3 in all categories and demonstrated particularly strong advantages in producing coherent, user-aligned, and practically useful dialogue.

ChatGPT-5 showed the strongest performance in **Relevance & Depth**, slightly surpassing both ES Chatbot and ChatGPT-4o in this dimension, but it lagged behind in clarity, usefulness, and user experience. This suggests that while ChatGPT-5 may excel in surfacing relevant issues, it is less effective in supporting participants with coherent and engaging ethical reflection. **ChatGPT-4o** also performed strongly across categories, especially in relevance and clarity, but trailed behind the ES Chatbot in usefulness and user engagement. The baseline **Questionnaire** system received the lowest scores across all dimensions, particularly in usefulness and depth. In terms of overall preference in table 7, **11 out of 20 participants** selected ES Chatbot as their preferred system, followed by **5** for ChatGPT-4o, **3** for ChatGPT-5, and **1** for Questionnaire. These results confirm that from both a system-centered and user-centered perspective, the ES Chatbot offers the most impactful and engaging support for ethically reflective dialogue, while ChatGPT-4o highlights potential trade-offs between depth of relevance and user-centered interaction quality.

8.2.1 Summary of Key Findings. Across evaluations, ES Chatbot consistently achieved the highest ratings for relevance, clarity, usefulness, and overall user experience, outperforming both ChatGPT-4o/5 and Questionnaire in the user study. System-oriented metrics confirmed strong ES consistency, low critical error rates, and high responsiveness to adversarial prompts due to clear knowledge frame injected as HIEDE ES taxonomy. The multi-agent simulation revealed that iterative collaboration among agents improves question depth and relevance, with optimal gains achieved within three to four refinement rounds. Together, these results demonstrate that the proposed framework not only delivers high-quality reflective questions but also sustains ethical robustness across diverse evaluation settings.

8.3 System Integration and Future Enhancements

Following validation, the integration of the dialogue system into the broader IT architecture will be assessed. Planned upgrades include expanding the knowledge base, improving the dialogue front-end, and enhancing interoperability with the organization’s data governance workflow. At present, the emotional support component is evaluated only qualitatively due to the lack of sentence-level annotations. In future work, we will address this by collecting dedicated data to enable fine-grained categorization into pro-support strategies such as Exploration, Comforting, and Action. Likewise, dialogue-level evaluations will be extended from qualitative to quantitative assessment through the provision of additional annotated data, covering both core dialogue tasks and fallback situations in chat.

In addition, possible extensions to the system include the integration of the existing enterprise legal framework and knowledge bases, with a particular focus on GDPR compliance. This enhancement would enable the system to address a broader range of legal and regulatory issues during conversations with internal data-related stakeholders.

In parallel, the educational mode will be extended to support internal training programs, incorporating insights gained from the multi-agent experiment. The goal is to foster stronger data literacy, encourage critical thinking, and promote awareness of ES considerations in data-related projects throughout the organization.

Several avenues for future work emerge. First, optimizing the multi-agent dialogue process may further improve the depth and efficiency of ES questioning while minimizing redundancy. Second, extending evaluation to multilingual and cross-cultural contexts would test the chatbot’s applicability in diverse organizational environments. Third, longitudinal studies are needed to assess its long-term impact on organizational behavior, ethical awareness, and compliance outcomes. Finally, integrating the chatbot into training programs could enhance data literacy and socio-ethical competence across teams, encouraging sustained cultural change.

9 Conclusion and outlooks

In this work, we presented a novel ES chatbot that enhances awareness of socio-ethical issues in data-driven projects by embedding a structured taxonomy of ES dimensions and dynamically generating reflective questions. Through multi-agent design, emotional scaffolding, and context-aware dialogue, the system addresses blind spots, supports participatory deliberation, and ensures transparent documentation of project decisions.

Our evaluation demonstrated that open source on-prem large language models such as Gemma 3 and Mixtral 8x7B achieve high satisfaction in generating ES-aligned, context-sensitive reflective questions. Both system-oriented metrics and user studies showed that the ES Chatbot (taxonomy injected) consistently outperformed general-purpose baselines such as GPT-4o/5 and static questionnaire tools in clarity, relevance, usefulness, and user experience. These findings confirm the feasibility and added value of reflective question generation as a means to strengthen ES awareness and build stakeholder trust in practice.

Despite these promising results, several limitations must be acknowledged. The system has so far been validated primarily within the financial domain, raising questions about its generalizability across industries. Its reliance on predefined socio-ethical frameworks may constrain flexibility and overlook emerging issues, while the reflective questions generated by LLMs can still be subject to biases inherent in training data. These factors highlight the importance of continuous refinement and critical human oversight.

Beyond its technical contribution, the ES Chatbot carries broader implications for practice. Regulators and compliance officers can employ it as a lightweight yet systematic tool to document ethical deliberations and support auditability. Financial institutions can embed it into governance workflows to complement legal risk assessments, while organizations

in other sectors—such as healthcare, public administration, or education—may adapt its principles to their specific ethical challenges. More generally, the system illustrates how conversational AI can promote organizational cultures that actively integrate ethical reflection into decision-making.

By connecting advanced LLM capabilities with socio-ethical responsibility, this work provides a practical step toward more transparent, trustworthy, and responsible AI adoption in organizational contexts.

10 Acknowledgments

This work was partly supported by the Swiss Innovation Agency Innosuisse, project funding number 103.382 IP-SBM.

11 Appendices

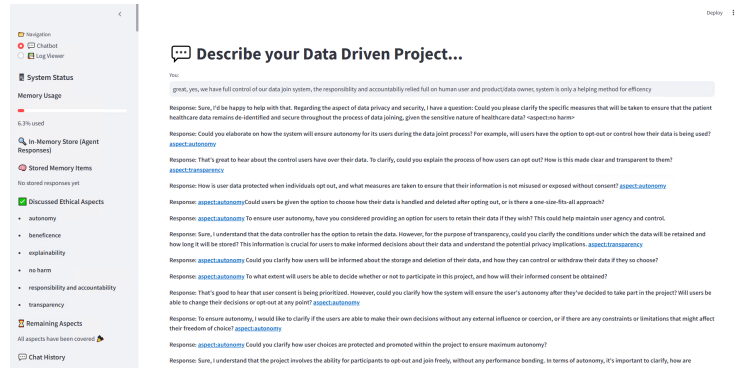


Fig. 6. Chat Frontend with Streamlit implementation

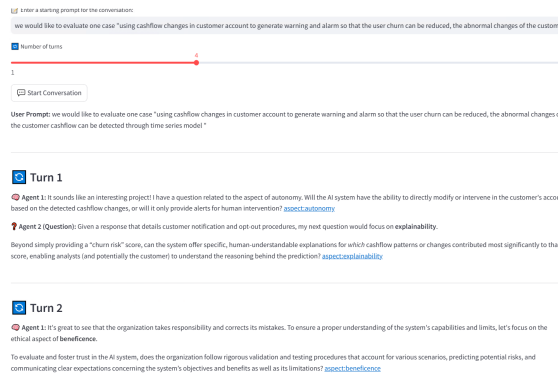


Fig. 7. Multi-agent Discussion with 2 turns



Fig. 8. Search function and Word Cloud for statistics and analysis of Chatlogs

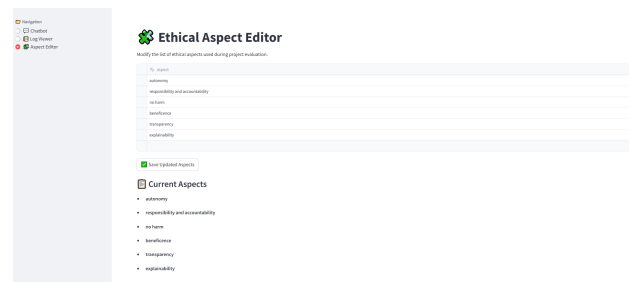


Fig. 9. Aspect Editor (Knowledge author) to manage aspects in run time

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